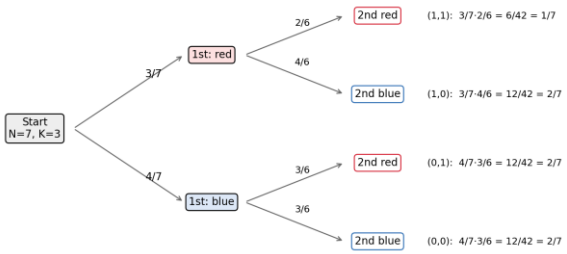
Summary card — Topic 5: Two Draws Without Replacement

Statistics · Midterm Team Project: Joint Distributions

1. Model	Topic 5 — Two draws without replacement. Urn with N balls, K red. X_1, X_2 = 1 if the 1st / 2nd ball is red, else 0. $p = K/N$, $q = 1 - p$.
2. Joint pmf and support	Support $\{0,1\}^2$ (a rectangle). $N(N-1)$ equally likely ordered pairs. $p(1,1) = K(K-1)/[N(N-1)]$ $p(1,0) = p(0,1) = K(N-K)/[N(N-1)]$ $p(0,0) = (N-K)(N-K-1)/[N(N-1)]$ Valid: all ≥ 0 and the sum $= [K(K-1) + 2K(N-K) + (N-K)(N-K-1)] / [N(N-1)] = 1$.
3. Marginals	Sum the joint pmf: $P(X_1=1) = P(X_2=1) = K/N$, so $X_1, X_2 \sim$ Bernoulli(p). The 2nd ball has the same marginal as the 1st although the draws are dependent.
4. Conditionals	$P(X_2=1 \mid X_1=1) = (K-1)/(N-1)$ $P(X_2=1 \mid X_1=0) = K/(N-1)$ $(K-1)/(N-1) < K/N < K/(N-1)$ Same formulas for X_1 given X_2 (the joint pmf is symmetric).
5. E, Var, Cov, ρ	$E(X_i) = p$; $Var(X_i) = pq$; $E(X_1X_2) = p(1,1) = K(K-1)/[N(N-1)]$ $Cov(X_1,X_2) = -K(N-K) / [N^2(N-1)] = -pq/(N-1)$ $\rho = -1/(N-1)$ (does not depend on K) Negative: a red first ball removes one red ball. $\rho \to 0$ when $N \to \infty$. $E(X_2 \mid X_1=x) = (K-x)/(N-1)$.
6. Independence	Criterion: $p(x_1,x_2) = p(x_1)p(x_2)$ for all cells and a rectangular support. At (1,1): $p(1,1) - (K/N)^2 = -K(N-K)/[N^2(N-1)] \neq 0$, so dependent (for $0<K<N$). But exchangeable: $p(1,0) = p(0,1)$, i.e. $p(x_1,x_2) = p(x_2,x_1)$.
7. Sum $S = X_1 + X_2$	$S \sim$ Hypergeometric(N, K, 2): $P(S=s) = C(K,s)\cdot C(N-K,2-s) / C(N,2)$, $s = 0, 1, 2$ ($P_0 = (N-K)(N-K-1)/[N(N-1)]$, $P_1 = 2K(N-K)/[N(N-1)]$, $P_2 = K(K-1)/[N(N-1)]$). $E(S) = 2p$; $Var(S) = 2pq - 2pq/(N-1) = 2pq(N-2)/(N-1)$ — equals $n\cdot pq\cdot(N-n)/(N-1)$ with $n = 2$.
8. Main Part D result	For n draws, $S_n = I_1+\ldots+I_n$, $Cov(I_i,I_j) = -pq/(N-1)$ for each of the $n(n-1)/2$ pairs, so $Var(S_n) = npq \cdot (N-n)/(N-1)$ (finite-population correction). Check: $n = N$ gives 0. With replacement Var = npq. If $N \to \infty$ and $K/N \to p$, Hypergeometric(N,K,n) → Binomial(n, p).
9. Picture	 Probability tree for N = 7, K = 3: p(1,1) = 1/7, the other three cells = 2/7.
10. Real example	Inspect 2 of 10 chargers, 3 defective: P(both defective) = (3/10)(2/9) = 1/15 ≈ 0.067; P(at least one defective) = 8/15 ≈ 0.533.